ET4351 Digital VLSI Systems on Chip

Project Grading Rubric

Academic Year 2025–2026

Criteria	9–10 (Excellent)	7–8 (Good)	6 (Sufficient)	0–5 (Insufficient)
Section 1: Architecture and Design Methodology				
SoC Digital HW/SW Architecture (30%)	• The design specifications are clear (quantitative and exhaustive). • The proposed hardware/algorithm co-design approach is clearly justified and provides an overview of the chosen algorithm at the software level, which bottleneck has been identified and why (with numbers), and thus which part is offloaded to a custom hardware accelerator. • Design choices for the accelerator are sound and properly justified. • IP selection for the memories, if different from the baseline, is justified from a power-performance-area point of view (quantitative). The memory hierarchy (i.e. use of SRAM and/or registers) is discussed. • Clean diagrams are provided for the resulting system-on-chip architecture, the memory map, and the accelerator architecture. • The resulting system shows a creative hardware/algorithm optimization that achieves an excellent power-performance-area tradeoff optimization.	• The design specifications are clear (quantitative and exhaustive). • The proposed hardware/algorithm co-design approach is clearly justified and provides an overview of the chosen algorithm at the software level, which bottleneck has been identified and why (with numbers), and thus which part is offloaded to a custom hardware accelerator. • Design choices for the accelerator are sound and properly justified. • IP selection for the memories, if different from the baseline, is justified from a power-performance-area point of view (qualitative). • Clean diagrams are provided for the resulting system-on-chip architecture, the memory map, and the accelerator architecture.	• Most of the design specifications are clear (quantitative and exhaustive). • A well identified part of a pathfinding algorithm is carried out by a dedicated hardware accelerator. • Clean diagrams are provided for the resulting system-on-chip architecture and the accelerator architecture.	The project guidelines are not met (for example: specifications missing, pre-routing, no hardware acceleration) or the design does not meet quality expectations (obvious flaw in the design, etc.).
Section 2: Implementation				
Design-Target-Driven Use of the Tools (20%)	• The optimization options and constraints of the design tools have been creatively and rigorously set to the benefit of the design target optimization. This is documented clearly and concisely in the report, without diving into implementation details. • The floorplan of the design is sound and its impact on congestion is discussed.	• The options/constraints set for the tools is relevant and summarized clearly. • The floorplan of the design is sound and its impact on congestion is discussed.	Some relevant options/constraints are explained, but miss relevant aspects and/or are lost among implementation details. The approached followed for the floorplan is explained.	There is no information as to how the tools were appropriately used and configured to support optimization of the design targets.
Signoff Verification (20%)	• The report convincingly shows that the final post-layout design is timing-clean, and that it is free from DRC and DRV violations (except for max_tran violations that are outside the scope of this course). • All reports are available in appendix.	• The design has reached the post-layout stage and is timing-clean. The report shows that, with minor rework, the design can be made clean (e.g., <3 DRC errors except for max_tran). • Proper justification and interpretation is given for the remaining errors. • All reports are available in appendix.	• The design has reached the post-layout stage and the reports show that, with some rework, the design can be made clean (e.g., <10 DRC errors except for max_tran, minor timing violations). • All reports are available in appendix.	The design has not reached the post-layout stage, or reports are missing or show multiple timing violations and/or DRC violations.
Section 3: Performance Metrics and Validation				
Performance Metric Extraction (15%)	• The power-performance-area metrics are extracted rigorously (quantitative and exhaustive), and demonstrate excellent performance levels. • For power/energy metrics, activity annotation has been performed correctly and is demonstrated by $\sim$100% annotation coverage. • Activity annotation report is provided in appendix, as well as full power report.	• Performance metrics are chosen and extracted properly, while activity annotation has been performed correctly with high coverage. • The design demonstrates convincing performance levels. • Activity annotation report is provided in appendix, as well as full power report.	• Performance metrics are present but some are lacking or missing rigor. • Activity annotation has been performed but cannot be fully validated.	Performance metrics are not extracted or not in a reliable manner.
Quality of Post-Layout Simulations (15%)	• The accelerator and SoC-level functional specifications of the proposed design are used to extract an exhaustive list of elements to be validated with post-layout simulation (i.e. test specifications). • The post-layout simulation waves are readable and clear (zoom level, chosen signals, comments and highlights). • The part on which activity annotation has been performed is highlighted.	• Test specifications and post-layout simulation waves allow for a solid validation of the design, with only minor issues in terms of clarity or missing tests. • The part on which activity annotation has been performed is highlighted.	• Test specifications are partially there, and the post-layout simulation waves allow validating key parts of the design, although with some clarity/exhaustivity issues.	Test specifications are mostly absent and post-layout simulation waves, if any, are unclear and/or do not allow properly validating the design.

Knock-Out Criteria	• The report fits within a **6-page limit** (excl. cover page and appendices; 10 pt font, IEEE double-column format), uses the mandatory section structure (Sections I–IV), includes a short introduction, and was submitted on time (Friday, April 10, 2026 at 16:59 CET). • Both a High-Performance (HP) and an Energy-Efficient (EE) accelerator design are presented, with `finaldesign_hp/` and `finaldesign_ee/` directories each containing `accel_audio.hex`, `et4351.phys.sdf`, and `et4351.phys.v`. • SystemVerilog/Verilog source files and all synthesis and P&R scripts are provided and cleaned for straightforward reading by an external evaluator. • All signoff reports, activity annotation reports, and full power reports are provided both as files in the project zip **and** as screenshots/excerpts in the PDF appendix. • Proper use of technical terms and reasonably proofread report with straightforward understanding. Clean, readable diagrams.